Problem 1

a. Additive Identity: $[(0,1)]$

Multiplicative Identity: $[(1,1)]$

b. ***Addition***

$[(p,q)]+[(a,b)]\equiv[(bp+aq,\ bq)]$

We need to check that this is well defined. Let $[(p,q)]\sim[(p_1,q_1)]$ and $[(a,b)]\sim[(a_1,b_1)]$. This gives $pq_1=qp_1$ and $ab_1=ba_1$. We wish to show that $[(p,q)]+[(a,b)]\sim[(p_1,q_1)]+[(a_1,b_1)]$. By definition, this requires that

$$[(pb+aq,qb)]\sim[(p_1b_1+a_1q_1,q_1b_1)]\Rightarrow q_1b_1(pb+aq)=qb(p_1b_1+a_1q_1).$$

We have

$$\begin{aligned}q_1b_1(pb+aq)&=q_1b_1pb+q_1b_1aq\\&=(pq_1)b_1b+(ab_1)q_1q\\&=(qp_1)b_1b+(ba_1)q_1q\\&=qb(p_1b_1)+qb(a_1q_1)\\&=qb(p_1b_1+a_1q_1)\end{aligned}$$

Multiplication

$[(p,q)]\cdot[(a,b)]\equiv[(ap,\ bq)]$

We need to check that this is well defined. Let $[(p,q)]\sim[(p_1,q_1)]$ and $[(a,b)]\sim[(a_1,b_1)]$. This gives $pq_1=qp_1$ and $ab_1=ba_1$. We wish to show that $[(p,q)]\cdot[(a,b)]\sim[(p_1,q_1)]\cdot[(a_1,b_1)]$. By definition, this requires that

$$[(ap,bq)]\sim[(a_1p_1,b_1q_1)]\Rightarrow apb_1q_1=bqa_1p_1.$$

We have

$$\begin{aligned}apb_1q_1&=(ab_1)(pq_1)\\&=(ba_1)(qp_1)\\&=bqa_1p_1\end{aligned}$$

Additive Inverse

$-[(p,q)]\equiv[(-p,q)]$

We need to check that this is well defined. Let $[(p,q)]\sim[(p_1,q_1)]$, so $pq_1=qp_1$. We wish to show that $-[(p,q)]\sim-[(p_1,q_1)]$. By definition, this requires that

$$[(-p,q)]\sim[(-p_1,q_1)]\Rightarrow -pq_1=q(-p_1).$$

We have

$$\begin{aligned}-pq_1&=-(pq_1)\\&=-(qp_1)\\&=q(-p_1)\end{aligned}$$

Multiplicative Inverse

$[(p,q)]^{-1} \equiv [(q,p)]$

We need to check that this is well defined. Let $[(p,q)] \sim [(p_1,q_1)]$, so $pq_1 = qp_1$. We wish to show that $[(p,q)]^{-1} \sim [(p_1,q_1)]^{-1}$. By definition, this requires that

$$[(q,p)] \sim [(q_1,p_1)] \Rightarrow qp_1 = pq_1 .$$

But this has already been established to be true.

c. ***Associativity of Addition***

$$([(p,q)]+[(r,s)])+[(t,u)]=[(ps+qr,qs)]+[(t,u)]=[((ps+qr)u+qst,qsu)]$$
$$=[(psu+qru+qst,qsu)]=[(psu+(ru+st)q,qsu)]=[(p,q)]+[(ru+st,su)]$$
$$=[(p,q)]+([(r,s)]+[(t,u)])$$

Commutativity of Addition

$$[(p,q)]+[(r,s)]=[(ps+qr,qs)]=[(rq+sp,sq)]=[(r,s)]+[(p,q)]$$

Additive Identity

$$[(0,1)]+[(p,q)]=[(0q+1p,\ 1q)]=[(0+p,\ q)]=[(p,q)]$$

Additive Inverse

$$[(p,q)]+(-[(p,q)])=[(p,q)]+[(-p,q)]=[(pq+q(-p),\ q^2)]=[(pq-pq,\ q^2)]=[(0,\ q^2)]$$

Then $0=0 \Rightarrow (0)(1)=(q^2)(0) \Rightarrow [(0,q^2)] \sim [(0,1)]$, thus $[(p,q)]+(-[(p,q)])=[(0,1)]$.

Associativity of Multiplication

$$([(p,q)]\cdot[(r,s)])\cdot[(t,u)]=[(pr,qs)]\cdot[(t,u)]=[((pr)t,(qs)u)]$$
$$=[(p(rt),q(su))]=[(p,q)]\cdot([(r,s)]\cdot[(t,u)])$$

Commutativity of Multiplication

$$[(p,q)]\cdot[(r,s)]=[(pr,qs)]=[(rp,sq)]=[(rp,sq)]=[(r,s)]\cdot[(p,q)]$$

Multiplicative Identity

$$[(1,1)]\cdot[(p,q)]=[(1p,1q)]=[(p,q)]$$

Multiplicative Inverse

$$[(p,q)]\cdot[(p,q)]=[(pq,qp)]=[(pq,pq)]$$

Then $1=1 \Rightarrow 1(pq)=1(pq) \Rightarrow [(1,1)] \sim [(pq,pq)]$, thus $[(p,q)]\cdot[(p,q)]=[(1,1)]$

Distributivity

$$[(p,q)]\cdot([(r,s)]+[(t,u)])=[(p,q)]\cdot[(ru+st,su)]=[(p(ru+st),qsu)]$$
$$=[(pru+pst,qsu)] \sim [(q(pru+pst),q(qsu))]=[(pqru+pqst,q^2su)]$$
$$=[(pr,qs)]+[(pt,qu)]=[(p,q)]\cdot[(r,s)]+[(p,q)]\cdot[(t,u)]$$

d. ***Property 1***

We would like to show that $[(p,q)] \le [(r,s)] \Rightarrow [(p,q)] + [(t,u)] \le [(r,s)] + [(t,u)]$.

$[(p,q)] \le [(r,s)] \Rightarrow ps \le qr \Rightarrow (ps)u \le (qr)u \Rightarrow psu + qst \le qru + qst \Rightarrow (pu+qt)s \le q(ru+st)$

$\Rightarrow (pu+qt)su \le qu(ru+st) \Rightarrow [(pu+qt, qu)] \le [(ru+st, su)]$

$\Rightarrow [(p,q)] + [(t,u)] \le [(r,s)] + [(t,u)]$. Note that $u \in \mathbb{N}$.

Property 2

We would like to show that $[(p,q)] \ge [(0,1)]$ and $[(r,s)] \ge [(0,1)] \Rightarrow [(p,q)] \cdot [(r,s)] \ge [(0,1)]$.

If $[(p,q)] \ge [(0,1)]$ and $[(r,s)] \ge [(0,1)]$, then $0(q) \le 1(p) \Rightarrow p \ge 0$ and $0(s) \le 1(r) \Rightarrow r \ge 0$.

Thus $pr \ge 0 \Rightarrow qs(0) \le pr(1) \Rightarrow [(pr, qs)] \ge [(0,1)] \Rightarrow [(p,q)] \cdot [(r,s)] \ge [(0,1)]$.

e. Let us assume that there exists a rational number such that $[(p,q)] \cdot [(p,q)] = [(p^2, q^2)] \sim [(2,1)]$. Then $p^2 = 2q^2$, and so p^2 is even, meaning p is even, since if it were not, p^2 would be odd. Thus $p = 2p_1$ for some $p_1 < p$. But then $2q^2 = (2p_1)^2 = 4p_1^2 \Rightarrow q^2 = 2p_1^2$, and by similar arguments, we find that $q = 2q_1$ for some $q_1 < q$. If we continue this argument, we will find that there will always be a number $p_{n+1} < p_n$ and a number $q_{n+1} < q_n$. Since $p_n, q_n \in \mathbb{N}$, this is a contradiction due to the fact that the natural numbers are bounded below, and therefore the sequences must terminate.